



THESIS ECONOMETRICS AND DATA SCIENCE

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# Evaluating the RVU levy exemption using AI-generated survey responses

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19 February 2026

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## Abstract

This thesis evaluates the RVU levy exemption, a Dutch regulation designed to help employees with demanding jobs retire up to 3 years early. As data on demanding work is not readily available, we use machine learning to predict whether exemption users had demanding work before they retired. To this end, we select six questions from the LISS Work & Schooling survey related to different aspects associated with demanding work. We train a machine learning model using individual-level information from the CBS microdata to predict what the response to each question would have been. Using these predictions, we assess the fraction of exemption users who had demanding work before retiring, and whether they are over- or underrepresented compared to the employed population and the exemption-eligible population. Our analysis suggests that individuals with demanding jobs make up a significant share of exemption users and are overrepresented compared to both the employed population and the exemption-eligible population. However, this analysis requires the unlikely assumption of nondifferential prediction errors, the validity of which strongly influences the results, as shown by a robustness analysis. Hence, further research is required to confidently evaluate the RVU levy exemption.

# Acknowledgements

## SEO Amsterdam Economics

First, I would like to thank SEO Amsterdam Economics for the opportunity to write my thesis with them. Being able to conduct my research at an institution with this level of expertise and access to facilities such as the CBS microdata environment has been a unique and valuable opportunity, especially for a master's thesis. In particular, I would like to thank my first supervisor, Jellien Knol, for her valuable input, the laughs we shared, and the foosball games we played.

## LISS panel

In this paper, we make use of data from the LISS panel (Longitudinal Internet studies for the Social Sciences), which is administered and managed by the non-profit research institute Centerdata (Tilburg University, the Netherlands). The LISS panel data were collected by the non-profit research institute Centerdata (Tilburg University, the Netherlands). Funding for the panel's ongoing operations comes from the Domain Plan SSH and ODISSEI since 2019. The initial set-up of the LISS panel in 2007 was funded through the MESS project by the Netherlands Organization for Scientific Research (NWO).

## CBS

Results based on calculations by SEO Amsterdam Economics in project number 3081 using non-public microdata from Statistics Netherlands. Under certain conditions, these microdata are accessible for statistical and scientific research. For further information: [Microdata: Conducting your own research | CBS](#).

## UWV

I would like to thank the Uitvoeringsinstituut Werknemersverzekeringen (UWV) for providing the dataset on RVU users and for allowing me to use it for my thesis, which was essential to this research.

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# 1 Introduction

## 1.1 Motivation and research question

To increase labour market participation among older citizens, the statutory retirement age (SRA) in the Netherlands, also referred to as the AOW age, has increased stepwise from 65 to 67 years in recent years. In the coming years, the SRA was set to increase by 8 months for every additional year in life expectancy. However, in January 2026, a new coalition agreed to increase the SRA more quickly than originally planned, to 12 months per additional year in life expectancy. In addition, early retirement policies have become stricter, giving fewer people the opportunity to retire before the SRA (Visser et al., 2016).

Where increasing the SRA reduces public costs and increases public income, concerns have been raised about the impact on people with demanding jobs. In the past, many already struggled to reach retirement in good health (Vermeer et al., 2016) and some entered social support programmes (Chirikos and Nestel, 1991). With future increases, this problem is likely to get worse.

To partially alleviate this issue, a new early retirement policy was introduced in 2021: the RVU levy exemption. This regulation offers the possibility of fiscal benefits when retiring up to 3 years before the SRA, and is aimed at supporting employees who are unable to reach the SRA in good health due to the heavy nature of their work (Ministerie van Sociale Zaken en Werkgelegenheid, 2025a). Since it was deemed impossible to determine at a central level what constitutes heavy work, the exemption was implemented as a general tax measure, and it was left to employers and employees to determine who can use an RVU, either through a collective or an individual agreement.

Originally, the exemption was scheduled to run until the end of 2025, but following the agreement "Gezond naar het Pensioen", it will be continued from 2026 onward, with the aim of becoming structural (Ministerie van Sociale Zaken en Werkgelegenheid, 2025c). The exemption will remain a general tax measure, but continuation is contingent on RVUs being used more targeted and controlled during CAO negotiations (Ministerie van Sociale Zaken en Werkgelegenheid, 2025a). Continuing from 2026, TNO will ex ante evaluate for CAO agreements whether they are in line with the intended goal, and evaluate annually how RVUs are used in CAO agreements.

Even though employers bear the main financial burden of an RVU, misuse of the exemption can cost the public substantial sums (toevoegen). Therefore, it is important to properly evaluate the exemption's efficiency, especially given the recently proposed increases to the SRA. However, research into its efficiency in the period is limited to two studies.

Ministerie van Sociale Zaken en Werkgelegenheid (2025b) investigates characteristics of RVU users in the period 2021-2024 and shows that in 2024, male users and slightly higher middle incomes are overrepresented. Ministerie van Sociale Zaken en Werkgelegenheid (2025d) investigates

requirements for RVUs as agreed in CAOs in January 2025, and shows that out of 273 CAOs with a collective RVU, 36 were fully generic and 63 only required a few years of service. Moreover, almost no CAOs include objective criteria for what is classified as demanding work.

Where these studies do provide some insights into whether the exemption worked as intended, the former do not connect to heavy work, and the latter do not examine actual RVU use but only the possibilities offered through collective agreements. Hence, to our knowledge, there is no available research that assesses the efficiency of the RVU levy exemption by investigating the relationship between RVU use and demanding work in the period 2021-2025. Therefore, this thesis aims to answer the question:

*How efficient is the RVU levy exemption to support people with demanding jobs with early retirement?*

The difficulty in answering this question is twofold. Firstly, there is no clear definition of what jobs are classified as "demanding", and there are different aspects that can make work demanding, physically or mentally (van den Bogaard et al., 2016). Secondly, even when agreeing on which aspects make a job demanding, there is no data available about these aspects for the full population.

## 1.2 Procedure and problems

To evaluate the exemption, we will compare the group of exemption users to 1) the employed population, represented by a random sample of individuals who appear in the Dutch employment register, and 2) the exemption-eligible population, represented by a sample of people who appear in the register with the same age distribution as exemption users.

To account for the lack of data on demanding jobs, we use a machine learning model to predict whether individuals had demanding work. Since the exemption came into effect in 2021, all users should have been part of the employed population the year before, and hence, predictions are made for 2020. As predictors, we use individual-level data from the CBS microdata, including education level, gender, and job-related data such as salary, extra hours worked, and job sector.

To account for the absence of a clear, general definition of what jobs are classified as demanding, several aspects of demanding work are considered. In particular, as a proxy for demanding work, we use responses to questions such as "Did/do you work irregular hours?" and "Did/do you need to lift heavy objects?" from the Longitudinal Internet Studies for the Social Sciences (LISS) panel Work & Schooling survey (Scherpenzeel and Das, 2010).

By evaluating the relationship between exemption use and responses to these questions, we can gain insights into which aspects of demanding work are associated with exemption use and whether the intended audience is reached. Moreover, we are mainly interested in the fraction of users who had jobs associated with each aspect of demanding work and whether they are

over- or underrepresented compared to the employed and exemption-eligible populations. We hypothesise that examining these relationships will provide insights into who used the exemption and whether the intended goals have been met.

A problem with this procedure is that the prediction model is unlikely to produce correct labels for everyone. In other words, our proxies for demanding work are likely to be subject to some degree of misclassification. There are methods that account for this, but to yield correct results, knowledge of misclassification rates is required. Under the assumption that misclassification rates are nondifferential between survey respondents and exemption users, they can be appropriately estimated on the sample of survey respondents. However, as we argue in this thesis, this assumption is unlikely to hold in practice.

### 1.3 Structure of this thesis

The remainder of this thesis is structured as follows: Section 2 covers the RVU levy exemption and problems that arise around classifying jobs as "demanding" more in-depth. Section 3 formalises the problem at hand and discusses theoretical prerequisites. Section 4 describes the available data and performs some preliminary analyses. Section 5 discusses the prediction model and its performance. Section 6 uses the generated predictions to investigate what claims can be made about the performance of the exemption and how robust these claims are. Section 7 concludes.

## 2 Background

To better understand the situation at hand, this section introduces the RVU levy exemption in more detail, followed by a more thorough description of the proposed changes in the "Gezond naar het Pensioen" agreement. Lastly, we discuss which jobs are classified as demanding and the assessment framework developed by TNO.

### 2.1 The RVU levy exemption

In 2021, a new regulation introduced the possibility of tax benefits for retiring up to 3 years early. This regulation, the RVU levy exemption, was specifically aimed at helping employees with demanding jobs retire early ([Ministerie van Sociale Zaken en Werkgelegenheid, 2025a](#)). We will introduce the exemption step by step by answering three questions.

#### What is an RVU?

An early retirement scheme (Regeling voor Vervroegde Uittreding (RVU), in Dutch) is an arrangement intended to bridge the financial gap between work and the SRA. That is, the employer pays the employee from when they stop working until the SRA, either at once or periodically. The arrangement can be made either individually between an employer and an

employee, or collectively through a collectieve arbeidsovereenkomst (CAO). The conditions for qualification as RVU, judged on objective conditions and features, rather than the motives of either the employer or employee, are twofold: 1) The dismissal has to be age-related, not dysfunction, reorganisation, or similar, and 2) the payment has to be large enough such that it can sustain the employee until retirement, determined by the 70%-toets. Details of the requirements can be found in [Ministerie van Sociale Zaken en Werkgelegenheid \(2025a\)](#).

### **What is the RVU levy?**

In a regular situation, when an RVU is agreed upon, a pseudo final levy has to be paid over the agreed amount. The applicable levy rate was 52% in 2025 but will gradually increase to 65% in 2028. This levy, known as the RVU levy, is meant to discourage people from agreeing on an RVU and keep people in the workforce longer ([Ministerie van Sociale Zaken en Werkgelegenheid, 2025a](#)).

### **What is the RVU levy exemption?**

The exemption threshold Early Retirement Scheme (drempelvrijstelling RVU-heffing), or RVU levy exemption, is a temporary relaxation of the RVU levy introduced on January 1st, 2021. Here, "threshold exemption" refers to the fact that only the amount below a certain threshold is exempt from the levy. In this case, the levy does not have to be paid over amounts up to the equivalent of the net AOW amount (€2273 in 2025 and €2,657 in 2026) for at most 36 months.

## **2.2 Gezond naar het Pensioen**

Originally, the exemption was set to last until the end of 2025, but following the agreement "Gezond naar het Pensioen", agreed on in 2024 ([Ministerie van Sociale Zaken en Werkgelegenheid, 2024](#)) and presented in 2025 ([Ministerie van Sociale Zaken en Werkgelegenheid, 2025c](#)), it has been extended from 2026 onwards with the aim to become structural and more targeted.

During the period 2021-2025, there were no clear guidelines on which jobs are considered demanding. [Ministerie van Sociale Zaken en Werkgelegenheid \(2025d\)](#) shows that this allowed for the presence of generic schemes that are not targeted at employees with demanding work. Moreover, almost no CAOs demarcated RVU use based on objective criteria. Therefore, for the upcoming year, RVUs should be used more targeted and based on objective criteria. Additionally, an important target in the upcoming years will be sustainable employability. That is, the aim should not only be to support employees who can no longer perform their work, but also to help them transition to gradually less demanding work as they get older.

Hence, for the upcoming years, an independent third party, TNO, has developed an assessment framework for what would be considered demanding in specific sectors. The new "TNO Expertisecentrum Zwaar Werk" will validate whether descriptions of RVU-eligible positions in new collective agreements align with these objective criteria.

Based on these criteria, discussed in the next subsection, CAO parties must deliver documentation



for each RVU-eligible function, including a description of the work, which load fields are part of the job, whether they pose a health risk, and whether they are avoidable. Based on this, TNO will validate whether the function is indeed demanding and provide advice. However, social partners remain responsible for determining who is eligible for an RVU.

The exemption will be subject to annual reporting and will be evaluated in 2028 and 2031 to determine whether the intended goals are met or a redirection is needed.

## 2.3 Demanding jobs

The discussion about which jobs are considered demanding is still ongoing. [Bakker and Demerouti \(2007\)](#) defines demanding jobs as "jobs that require sustained physical and/or psychological effort". [Hooftman and Houtman \(2010\)](#) and [van den Bogaard et al. \(2016\)](#) argue that psychologically stressful and psychosocially demanding jobs should also be taken into account, not just physically demanding ones. The Federatie Nederlandse Vakbeweging (FNV) argues there are three types of demanding jobs: 1) Physically demanding jobs that require heavy lifting and repetitive movements, 2) work that requires irregular work hours and night shifts, and 3) mentally demanding work that is emotionally heavy or with verbal abuse ([Federatie Nederlandse Vakbeweging, 2024](#)).

In 2025, TNO presented a new assessment framework to more objectively classify what jobs are demanding ([TNO, 2025](#)), specifically designed to validate whether descriptions of RVU-eligible positions in CAOs are part of the target audience. They specify the following criteria: the work must be demanding and pose health risks, the health risks must be inherent to the function and unavoidable, and expertise has been used to classify RVU-eligible functions.

Additionally, they provide a basic list with aspects that relate to demanding work, spread across 5 load fields:

- Working Times: either during the night, very early or very late, or on-call shifts.
- Psychosocial Load: socially unsafe, emotionally taxing, functional or social support.
- Cognitive Load: attention, perception capacity, decision-making ability, memory, time pressure
- Physical Load: lifting and carrying, pushing and pulling, unfavourable postures/movements, hand-arm tasks, vibrations and shocks
- Environmental Load: dangerous conditions, biological agents, radiation, heat/cold load, noise

They specifically mention that this list is not exhaustive, and some aspects only satisfy in combination with others. Moreover, they must pose a risk to the employee or others.

### 3 Theoretical framework

This section formalises the problem at hand and introduces the theoretical prerequisites of our approach. First, we formalise how we evaluate the efficiency of the exemption. Then, we cover some basic misclassification theory and how to correct for it when misclassification rates are known. We continue by discussing how predictions can lead to misclassification and how misclassification rates are estimated. Lastly, as the true responses of exemption users are not known, we show which assumptions are needed to estimate these on the available data. We show why the required assumption of nondifferentiality is unlikely to hold in practice. We argue for the nondifferentiality condition to hold and argue why they are highly unlikely.

#### 3.1 Entities of interest

Let  $Y$  be a binary variable denoting whether someone made use of the RVU levy exemption, and let  $X$  be a binary variable denoting whether someone had a demanding job. We are mainly interested in the probability that an exemption user had demanding work and whether individuals with demanding work are over- or underrepresented among exemption users.

Following the notation of [Buonaccorsi \(2010\)](#), the former is evaluated by looking at

$$\alpha_y = P(X = 1 \mid Y = y) = 1 - P(X = 0 \mid Y = y),$$

and the latter is evaluated by

$$P(X = 1 \mid Y = 1) \text{ vs } P(X = 1 \mid Y = 0),$$

which is assessed by estimating the odds ratio

$$\Psi = \frac{a_1(1 - a_0)}{a_0(1 - a_1)}, \tag{1}$$

where  $\Psi > 1$  indicates  $a_1 > a_0$ . The odds ratio is symmetric in the sense that switching the values of  $a_1$  and  $a_0$  is the same as inverting the odds ratio.

In our setting, we pre-stratify on exemption use and sample independently from the control group, consisting of individuals who did not use the exemption. This case-control setting simplifies comparisons with similar groups, but it prevents us from investigating the fraction of individuals with demanding work who use the exemption. Formally, this setting allows us to make statements about  $\alpha$ , as specified in Table 1e, and functions thereof, but it prevents us from doing the same about  $P(Y \mid X)$  ([Buonaccorsi, 2010](#)).

One might suggest that a regression of the form  $Y = \alpha + \beta X + \gamma Z$  would be appropriate to account for the effect of potentially relevant covariates  $Z$ , but we argue that the interest lies

more in population shares rather than the explanatory power of demanding work.<sup>1</sup>

Because exemption use is restricted only to employees in the last 3 years before the SRA, one might not only be interested in using the full employed population as a control sample, but also the population eligible for exemption use. As not all individuals make use of the full 3 years, some ages appear more frequently among exemption users. To account for this, we construct a comparison sample where each age is represented as frequently as in the exempted population.<sup>2</sup>

More formally, let  $C$  denote inclusion in a subsample such that  $P(\text{age} \mid Y = 0, C = 1) = P(\text{age} \mid Y = 1)$ , which indicates whether an individual is in the same age group as exemption users. Then we are also interested in

$$P(X = 1 \mid Y = 1, C = 1) \text{ vs } P(X = 1 \mid Y = 0, C = 1),$$

where  $P(X = 1 \mid Y = 1, C = 1) = P(X = 1 \mid Y = 1)$  holds trivially as all individuals that made use of the exemption ( $Y = 1$ ) are in that age group.  $P(X = 1 \mid Y = 0, C)$ , on the contrary, might be differential with respect to  $C$  as the probability of having demanding jobs might differ across ages.

### 3.2 The misclassification problem

The apparent problem is that there is no directly available data on demanding jobs for exemption users. That is, we know the number of people that made use of the exemption,  $n_{11} + n_{01}$ , but we do not know  $n_{11}$  and  $n_{01}$  themselves. Hence, we can not estimate  $\alpha_1$  directly. To estimate these conditional probabilities, for each individual, we generate predictions regarding whether they had demanding work, denoted by  $\hat{X}$ .

The imperfection of these predictions implies that our demanding work proxies are subject to misclassification. As a result,  $\hat{X} \neq X$  and hence  $P(Y \mid X) = P(Y \mid \hat{X})$  does not hold trivially. Therefore, naively estimating  $\alpha_y$  by

$$a_y = \frac{\dot{n}_{1y}}{\dot{n}_{1y} + \dot{n}_{0y}}, \quad (2)$$

where  $\dot{n}_{\hat{x},y}$  is the number of individuals with demanding work prediction  $\hat{x}$  and exemption usage  $y$ , can yield a biased estimate (Buonaccorsi, 2010). Not only can this lead to attenuation bias, but this can also lead to bias away from zero (Buonaccorsi, 2010).

<sup>1</sup>As an example, imagine a simplified and unrealistic situation where 90% of the individuals in the construction sector have demanding jobs and no individuals outside of the sector do so. Additionally, all individuals in the construction sector use the RVU, whereas none outside do. Then, estimating a regression with sector as a covariate would attribute all explanatory power to the sector and none to the demanding work variable, implying that the regulation does not work efficiently, whereas it reaches all individuals with demanding work and 90% of individuals who use the exemption have demanding work.

<sup>2</sup>In our study, we stratify on both gender and age. Severe computation times and limited time prevented us from rerunning the experiment with a different control sample. We consider this a limitation of our study.

|                |                |                |                           |                |                |                             |                  |                  |
|----------------|----------------|----------------|---------------------------|----------------|----------------|-----------------------------|------------------|------------------|
|                | $Y = 0$        | $Y = 1$        |                           | $Y = 0$        | $Y = 1$        |                             | $X = 0$          | $X = 1$          |
| $X = 0$        | $n_{00}$       | $n_{01}$       | $\hat{X} = 0$             | $\dot{n}_{00}$ | $\dot{n}_{01}$ | $\hat{X} = 0$               | $\ddot{n}_{00y}$ | $\ddot{n}_{01y}$ |
| $X = 1$        | $n_{10}$       | $n_{11}$       | $\hat{X} = 1$             | $\dot{n}_{10}$ | $\dot{n}_{11}$ | $\hat{X} = 1$               | $\ddot{n}_{10y}$ | $\ddot{n}_{11y}$ |
| (a) $n_{XY}$   |                |                | (b) $\dot{n}_{\hat{X},Y}$ |                |                | (c) $\ddot{n}_{\hat{X} XY}$ |                  |                  |
|                | $Y = 0$        | $Y = 1$        |                           | $Y = 0$        | $Y = 1$        |                             | $X = 0$          | $X = 1$          |
| $X = 0$        | $1 - \alpha_0$ | $1 - \alpha_1$ | $\hat{X} = 0$             | $1 - a_0$      | $1 - a_1$      | $\hat{X} = 0$               | $\theta_{0 0Y}$  | $\theta_{0 1Y}$  |
| $X = 1$        | $\alpha_0$     | $\alpha_1$     | $\hat{X} = 1$             | $a_0$          | $a_1$          | $\hat{X} = 1$               | $\theta_{1 0Y}$  | $\theta_{1 1Y}$  |
| (d) $P(X   Y)$ |                |                | (e) $P(\hat{X}   Y)$      |                |                | (f) $P(\hat{X}   XY)$       |                  |                  |

Table 1: An overview of counts and conditional probabilities relating to predictions  $\hat{X}$ , true states  $X$ , and outcomes  $Y$ .

There are several methods that appropriately account for measurement error, and the suitability of each depends on the available data and the problem at hand (see [Buonaccorsi \(2010\)](#) for an introduction and overview). In the context of this thesis, we are interested in the relationship between two categorical variables, exemption usage  $Y$  and aspects of demanding work  $X$ , where our measure of  $X$  may be misclassified. This is referred to as misclassification in two-way tables.

In this setting, correcting for misclassification is straightforward if misclassification probabilities are known. Following [Buonaccorsi \(2010\)](#), let  $\theta_{\hat{X}|X} = P(\hat{X} = \hat{x} | X = x)$  denote the probability of the demanding work prediction being equal to  $\hat{x} \in \{0, 1\}$ , given that the true response is, or rather would have been,  $x \in \{0, 1\}$ . Under possible misclassification, the probabilities of correct classification  $\theta_{1|1}$  and  $\theta_{0|0}$ , respectively referred to as the sensitivity ( $Se$ ) and specificity ( $Sp$ ), can be strictly smaller than one. Equivalently, the misclassification probabilities  $\theta_{1|0}$  and  $\theta_{0|1}$  might be nonzero.<sup>3</sup>

In addition, it is non-trivial that misclassification probabilities are equivalent between sub-populations. That is,  $\theta_{1|1}$  and  $\theta_{0|0}$  may differ between subsamples. For example, let  $\theta_{\hat{X}|XY} = P(\hat{X} = \hat{x} | X = x, Y = y)$  be the probability of classifying demanding work as  $\hat{x} \in \{0, 1\}$  given the true value  $x \in \{0, 1\}$  and exemption use  $y \in \{0, 1\}$ , then it could be that  $\theta_{\hat{X}|X1} \neq \theta_{\hat{X}|X0}$  and misclassification rates are differential with respect to  $Y$ . Here, they are differential in the sense that they differ between exemption users and non-exemption users.

When misclassification rates are known, we specify a corrected estimate of  $\alpha_y$  as

$$\hat{\alpha}_y = \frac{a_y - P(\hat{X} = 1 | X = 0, Y = y)}{P(\hat{X} = 1 | X = 1, Y = y) - P(\hat{X} = 1 | X = 0, Y = y)} = \frac{a_y - (1 - \theta_{0|0y})}{\theta_{1|1y} + \theta_{0|0y} - 1}, \quad (3)$$

where  $a_y$  is the naive estimate of  $\alpha_y$ , and  $\theta_{1|1y}$  and  $\theta_{0|0y}$  again denote the sensitivity and specificity,

<sup>3</sup>As we assume the true  $X$  is fixed, we are looking at misclassification. In a setting where the observed is fixed, following [Buonaccorsi \(2010\)](#) notation, the entities of interest are reclassification probabilities  $\lambda_{X|\hat{X},Y} = P(X = x | \hat{X} = x, Y = y)$ .

respectively, given  $Y = y$ . Section 3.4 briefly discusses the intuition behind this formula.

When the true responses are available, the sensitivity and specificity can be estimated by

$$\hat{\theta}_{i|iy} = \frac{\tilde{n}_{iiy}}{\tilde{n}_{iiy} + \tilde{n}_{(1-i)iy}}, \quad (4)$$

where  $\tilde{n}_{izy}$  denotes the number of individuals with prediction  $\hat{x} = i$ , true value  $x = z$ , and exemption use is  $Y = y$ .

For the population of survey respondents, the sensitivity and specificity can be adequately estimated by holding out a random test set on which the prediction model is not trained but only tested. However, as the true responses of exemption users and individuals in the control populations are generally unavailable, we cannot estimate them directly. As a result, assumptions have to be made to estimate misclassification rates for these populations.

Opposed to differentiability, nondifferentiability assumes that misclassification rates are equivalent between groups. In the situation of nondifferentiability of misclassification probabilities with respect to exemption usage, we can write

$$P(X | \hat{X}, Y) = P(X | \hat{X}), \quad (5)$$

or equivalent  $\theta_{\hat{x}|x,y} = \theta_{\hat{x}|x}$ .

Under the assumption of nondifferentiability of prediction errors, misclassification rates are transportable from the survey respondents to the population and can hence be estimated based on misclassification rates of survey respondents.

### 3.3 Misclassification due to prediction errors

In our setting, we predict whether individuals have demanding work based on the prediction model  $f(\cdot)$  and features  $Z$ . Moreover, the predicted values  $\hat{X} = \hat{f}(Z)$  are a deterministic function of the features. Training the prediction model amounts to learning about the conditional distribution  $P(X | Z)$  to determine which regions of  $Z$  map to  $\hat{X} = 0$  and which to  $\hat{X} = 1$ . That is, training  $f(\cdot)$  comes down to determining  $A_0$  and  $A_1$  such that

$$A_i = \{z \in \mathcal{Z} : f(z) = i \ \forall \ z \in A_i\}, \ i \in \{0, 1\}.$$

Since our model should predict well for both demanding and non-demanding work and we observe class imbalance, our definition of "optimal" is not simply the prediction accuracy  $P(\hat{x} = x)$ . Rather, we look at the underlying distribution  $P(\hat{X} | X)$  and aim to find  $f(\cdot)$  such that

$$A_i = \{z \in \mathcal{Z} : P(z | X = i) > P(z | X = 1 - i)\}, \ i \in 0, 1, \quad (6)$$

which maximises the mean of the sensitivity and specificity. This is attained by setting  $f(z)$  equal to the  $X$  that has the highest density for that  $z$ . A simplified visualisation of this can be found in Figure 1.

Given the trained predictor, predictions for new samples are made by predicting  $\hat{x} = 1$  if  $z \in \hat{A}_1$  and  $\hat{x} = 0$  otherwise. To avoid overestimating prediction performance due to overfitting, it is important to evaluate performance on an independent test set. The simplified visualisation in Figure 2 shows how prediction probabilities  $\theta$  can be estimated by calculating the fraction of each group that appears in  $\hat{A}_i, i \in \{0, 1\}$ . In this process, misclassification of predictions can occur for several reasons.

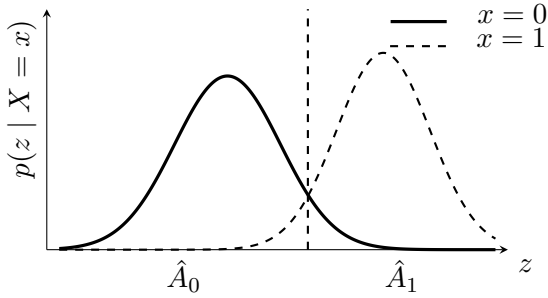


Figure 1: Training the predictor yields  $\hat{A}_0$  and  $\hat{A}_1$ .

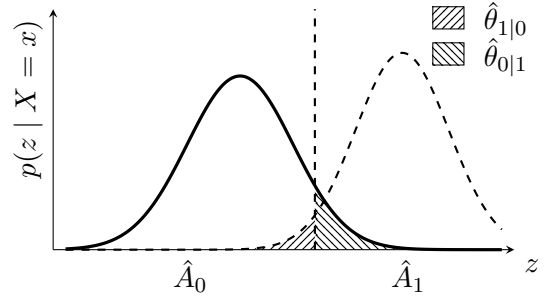


Figure 2: Evaluating performance on test set gives  $\hat{\theta}$ . The imperfection of the decision boundary is due to sampling, approximation, and estimation errors.

Unless the  $Z$  regions of  $X = 0$  and  $X = 1$  are completely separated, even the best possible deterministic prediction model of the form  $\hat{X} = f(Z)$  produces misclassified predictions. Here, misclassification probabilities depend on the fraction of the conditional density of  $Z | X = i$  that appears in the area where  $P(Z | X = i) < P(Z | X = 1 - i)$ . Formally, with  $A_i$  as in Eq. 6,

$$\hat{\theta}_{i|(1-i)} = \int_{z \in \mathcal{Z}} \mathbb{1}_{\{z \in A_{1-i}\}} P(z | X = i) = \int_{z \in A_{1-i}} P(z | X = i), \quad i \in \{0, 1\}. \quad (7)$$

Furthermore, the scenario where our trained model  $\hat{f}(\cdot)$  finds the truly optimal  $A_i, i \in \{0, 1\}$  is unlikely. As we do not have infinite data, the limited training data will not contain enough information to learn the optimal regions, resulting in estimation error. Additionally, due to sampling error, the training sample will not be a perfect representation of the full population. Lastly, some models suffer from approximation error due to restrictions on their functional form, which prevent them from forming optimal regions even with infinite data. However, this is more of a problem for traditional econometric models rather than recent, more flexible ML models. The effects resulting from a suboptimal decision region can also be seen in Figure 2.

Looking at Figure 2, it is easily explained why the formula for the corrected estimate of  $a_y$  in Eq. 3 makes sense intuitively. Using  $P(\hat{X} = 1 | X = 1) = \theta_{1|1}$  and  $P(\hat{X} = 0 | X = 0) = \theta_{0|0}$ , the

probability that an individual will be classified as  $\hat{X} = 1$  is equal to

$$P(\hat{X} = 1) = P(X = 1)\theta_{1|1} + P(X = 0)(1 - \theta_{0|0}), \quad (8)$$

which is a very intuitive formula. Rewriting this equation gives

$$P(X = 1) = \frac{P(\hat{X} = 1) - (1 - \theta_{0|0})}{\theta_{1|1} + \theta_{0|0} - 1}, \quad (9)$$

which corresponds to the corrected estimator in Eq. 3. As this explanation unveils again, an accurate approximation of the misclassification rates of predictions is needed for valid inference.

### 3.4 Nondifferentiability of prediction errors

For subsamples with known true values, the prediction performance can be accurately estimated by comparing predictions against these true values. Additionally, it is possible to estimate performance on a subsample of the test set, such as a specific age group. See Figure 3 for a simplified example of how prediction errors can differ when, for example, selecting a sample from a specific age cohort.

However, when making predictions for samples whose true values are unknown, estimating misclassification rates directly is not possible. As Figure 4 demonstrates, the fraction of predictions for each category is known, but directly estimating misclassification rates of the predictions requires either more data or assumptions. A straightforward assumption would be nondifferentiability of prediction errors.

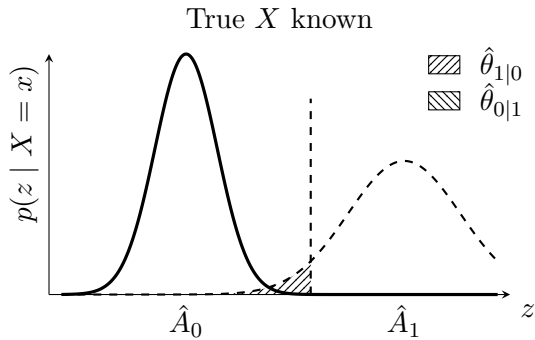


Figure 3: Evaluating performance for age-stratified sample gives alternative  $\hat{\theta}$ .

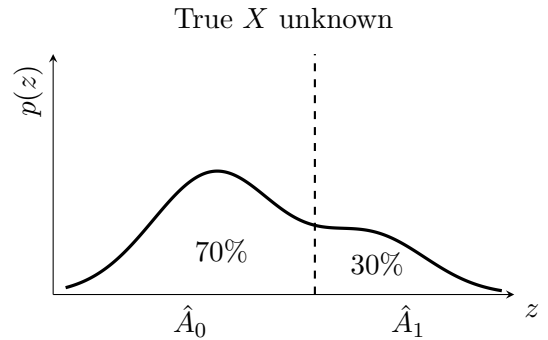


Figure 4: When true  $X$  is unknown,.

Combining Eq. 5 and Eq. 7, nondifferentiability of prediction errors requires between exemption users and non-exemption users,

$$\int_{z \in A_{1-i}} P(z | X = i, Y = 0) = \int_{z \in A_{1-i}} P(z | X = i, Y = 1), \quad i \in \{0, 1\}, \quad (10)$$

which says the density of  $z$  given  $x$  must have the same weight in regions  $A_i$  for  $Y = 0$  compared to  $Y = 1$ . This can be true in two ways: 1)  $P(Z | X, Y) = P(Z | X)$ , they have the same distribution, which is a sufficient condition for Eq. 10 to hold, or 2)  $P(Z | X, Y) \neq P(Z | X)$ , but the weight is distributed such that Eq. 10 still holds.

We argue that the former could hold structurally under restrictive assumptions, but the latter is very unlikely to hold structurally, as we hypothesise it could hold only reasonably by chance.

Namely, the former implies that, given demanding work, exemption usage is a random sample of individuals. In this case, differences in  $\hat{X}$  between  $Y = 0$  and  $Y = 1$  are only due to differences in  $P(X)$ . However, this assumption is also unlikely to hold, as we hypothesise that each individual with a demanding job did not have the same probability of using the exemption (for example, across specific sectors).

We can, however, be confident in ruling out some combinations of sensitivity and specificity. For example, in Figure 4, following Eq. 8, a specificity of 50% and a sensitivity of 100% would be highly unlikely to yield 30% of observations as  $\hat{x} = 1$ , as this combination would be expected to produce at least as many prediction for  $\hat{X} = 1$  for any  $P(X) \in [0, 1]$ .

Concluding, we hypothesise that the nondifferentiability assumption is unlikely to hold 1) because the exemption users could be sampled from a region of  $Z$  where performance is different from the comparison sample, and 2) because even given  $Z$ , RVU users could be a nonrandom sample with a different underlying relation (for example, due to unobserved variables).

## 4 Data

This thesis uses three data sources: 1) a list of individuals who made use of the RVU levy exemption, 2) responses to the LISS Work & Schooling survey, and 3) administrative data from the CBS microdata. All datasets were uploaded to/accessed in the CBS Microdata Remote Access (RA) environment.

### 4.1 CBS Microdata

The CBS microdata contains many detailed, anonymised individual-level datasets about the Netherlands. This ranges from personal records such as birth months and salaries to data on households and organisations. Use of this data is subject to strict confidentiality and is available only to approved research institutions.

In this research, we make use of the GBAPERSOONKTAB (Centraal Bureau voor Statistiek, 2021), SPOLISBUS (Centraal Bureau voor Statistiek, 2025), and HOOGSTEOPLTAB (Centraal Bureau voor Statistiek, 2019) microdatasets. The GBA (gemeentelijke basisadministratie) contains information on gender and birth year; the SPOLISBUS contains monthly data from



the "polisadministratie" that covers, among others, salary, hours worked, whether the individual had a car from the employer, sector, and extra hours worked. Lastly, the HOOGSTEOPLTAB contains the highest level of education achieved, which is classified as 1 (low), 2 (secondary), or 3 (high) ([Centraal Bureau voor Statistiek, 2026](#)).

## 4.2 RVU users

To define who is an exemption user, we use a list of individuals who have used an RVU. These individuals appear with code 53 (early retirement scheme) as a type of employment relationship in the SPOLISBUS. Since this data is not directly available in the regular SPOLISBUS, it was provided to us by the UWV. This dataset contains 27,016 individuals who used the RVU levy exemption.

## 4.3 LISS Work and Schooling survey

As proxies for demanding work, we make use of data from the LISS panel (Longitudinal Internet studies for the Social Sciences), which is administered and managed by the non-profit research institute Centerdata (Tilburg University, the Netherlands) ([Scherpenzeel and Das, 2010](#)). Particularly, we use responses to several questions from the LISS Work & Schooling survey. This yearly survey consists of 17 waves (2008 - 2024), and the number of participations per respondent ranges from 1 to 17. According to the LISS panel, their surveys are representative samples of the Dutch population ([Scherpenzeel and Das, 2010](#)).

Throughout the waves, questions have generally stayed the same or similar. However, sometimes questions were changed. From the survey, we identified six questions as most relevant regarding demanding work. These are,

- (a) **Profession:** What is your current profession / What profession did you exercise in your last job?
- (b) **Irregular hours:** Do/did you work irregular hours?
- (c) **Physically demanding (agree):** My job is/was physically demanding.
- (d) **Physically demanding (often):** Is/was your job physically demanding?
- (e) **Heavy lifting:** Do/did you need to lift heavy objects?
- (f) **Kneel / stoop:** Do/did you need to kneel or stoop?

where questions (b), (d), (e), and (f) were answered with never/sometimes/often, question (c) with strongly disagree/disagree/agree/strongly agree, and question (a) with nine answers, which we divided into the categories mental/manual/supervisory. We convert responses to these

questions into a binary indicator  $X$ , where  $X = 1$  indicates that work is demanding. Answers were classified as demanding if the response is often, agree/strongly agree, or supervisory.

Here, question (a), whether someone had a supervisory profession, might appear as a weird question. We include it as the preliminary analysis of the overlap between exemption users and LISS respondents showed they were strongly overrepresented in the overlap, compared to LISS respondents who did not take an RVU.

It is important to note that this survey consists of self-reported data and is, therefore, likely to contain measurement errors. However, we do not take this into account in the analysis. Out of 99,404 responses across different waves, 13,896 belonged to individuals who did not give permission to connect to their responses to the microdata, 1,768 to individuals who could not be found in the microdata, and 83,740 responses were connected.

#### 4.4 Comparison groups

To evaluate the RVU levy exemption, we want to compare the frequency of demanding jobs amongst exemption users to both the employed population and the exemption-eligible population. Therefore, we construct four comparison groups of employed individuals:

- (1) **Respondents:** respondents to the LISS Work & Schooling survey.
- (2) **Exempted:** people who used the RVU levy exemption.
- (3) **Employed:** a random sample of people who appear in the Dutch employment register in 2020 and did not use the exemption.
- (4) **Eligible:** a random sample of people who appear in the Dutch employment register in 2020, did not use the exemption, and have a similar age distribution to the exempted group.

A measure for demanding work is only available for respondents. For the other groups, predictions are made.

Groups (1) and (2) are predefined, as we can not influence participation in the LISS survey and exemption usage. A small sample of individuals who appear in the LISS survey are also exemption users. As they are assumed to be a random sample of (2), but not a random sample of (1), assigning them to (2) would induce a selection bias in (1). Hence, they are assigned to group (1).

Since the exemption can only be used by individuals in the employed population, groups (3) and (4) are constructed by randomly selecting individuals from the GBA who appeared in the SPOLIS 2020. For group (4), a weighted draw is performed, with weights proportional to the number of exemption users with the same birth year. With "eligible" we refer to the fact that they could have potentially used the exemption, even though they are not necessarily considered

to be part of the intended recipients. It should be noted that these two groups are a random sample of the employed non-user population rather than just the employed population. This subtle difference results in exemption users being underrepresented in comparison groups (3) and (4).

Table 2 shows the original sizes of each group and the number of individuals that appear in the 2010-2025 and 2020 editions of the SPOLIS, respectively. Remarkable is the number of individuals in the comparison group (2), as apparently not all individuals who used an RVU appeared in the SPOLIS between 2010 and 2025, and a significantly larger proportion did not appear in the 2020 SPOLIS.

| Group   | $n$     | $n_{\text{SPOLIS}}$ | $n_{\text{SPOLIS},2020}$ |
|---------|---------|---------------------|--------------------------|
| (1)-(4) | 200,055 | 129,019             | 93,362                   |
| (1)     | 13,059  | 10,441              | 7,745                    |
| (2)     | 27,016  | 27,006              | 26,618                   |
| (3)     | 100,000 | 51,969              | 31,483                   |
| (4)     | 60,000  | 39,623              | 27,535                   |

Table 2: Number of individuals that appear in employment register, by comparison group.  $n$  denotes the number of people who appear in the GBA.  $n_{\text{SPOLIS}}$  and  $n_{\text{SPOLIS},2020}$  denote the number of people that appear at least once between 2010 and 2025, and in 2020, respectively.

As argued in Section 3, inference validity depends on whether misclassification probabilities are nondifferential between comparison groups. In turn, nondifferentiability is influenced by whether the comparison groups are randomly sampled (given age) from the overall employed population.

According to the LISS panel, their respondents are a representative sample of the Dutch population. Hence, selecting only respondents who appear in the administrative records of a certain age should be a representative sample of that group. Additionally, selecting a random sample from the GBA and filtering it using the same criteria should yield a representative sample of the population as well. Therefore, if the LISS respondents are indeed a representative sample of the Dutch population, survey respondents and the two control groups (3) and (4) should be drawn independently from the same population.

As a result, following the reasoning of Section 3.4, the misclassification probabilities estimated on the test set should be transportable to the comparison sets, and valid inference could be performed. Section 4 investigates this claim. However, by assuming that exemption users are a random sample of the employed population, we implicitly assume they are a random sample of individuals with demanding jobs, which we think is unlikely to be true. As a result, we hypothesise that the prediction errors are not transportable to exemption users. Figures 5 and 6 show the distributions of several characteristics for the comparison groups.<sup>4</sup>

<sup>4</sup>To guarantee privacy, raw datasets can only be accessed in the secure RA environment, and results have to satisfy strict anti-disclosure criteria before they can be exported from the secure environment. One of these criteria

For comparison, survey respondents are filtered for appearance in the 2020 employment register, but to increase the sample size while training the ML model, individuals are kept if they appear at least once in the employment register between 2010 and 2025.

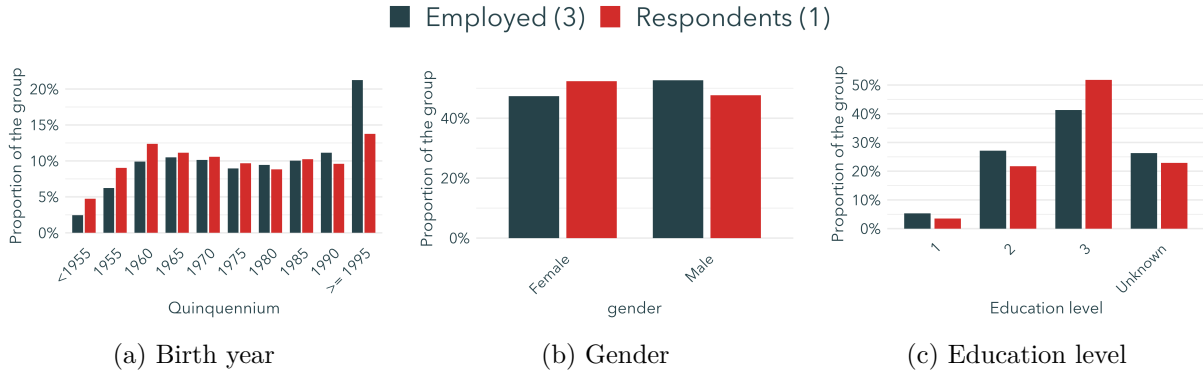


Figure 5: Characteristics distributions of respondents (1) and employed (3) groups.

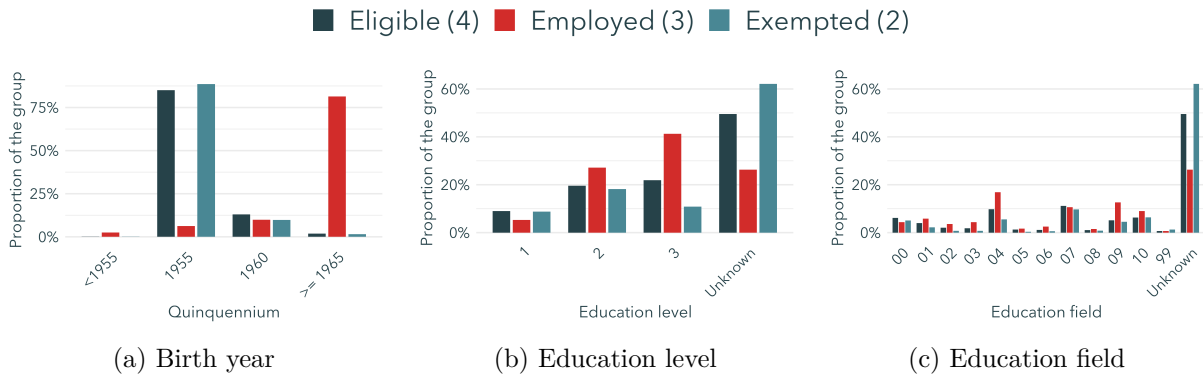


Figure 6: Characteristics distributions of the exempted (2), employed (3), and eligible (4) comparison groups.

Figure 5 shows the distribution of certain characteristics for the respondents (1) and employed (3) groups. The youngest age cohorts appear more often in the random sample, and the oldest appear more often in the respondents' sample. Also, females are slightly overrepresented in the LISS survey, as are individuals with a high level of education. These figures suggest that the respondents group (1) is not a representative sample of the employed population (3).

Figure 6 shows the distributions of several characteristics of the exempted (2), employed (3), and eligible (4) comparison groups. It is apparent that there are large differences between the random group (3) and groups (2) and (4). This makes sense as groups (2) and (4) consist of similar-aged individuals. Additionally, there are also some differences between groups (2) and

is that cells in tables must not be based on fewer than 10 individuals, so to comply with these standards, some results contain merged groups.

(4). This suggests that the exempted are not a representative sample of the exemption-eligible population, which aligns with our expectations.

The results suggest that neither the respondents (1) and employed (3), nor the eligible (4) and exempted (2) groups have a similar  $Z$ -distribution, and hence it is unlikely for the assumption of nondifferentiality to hold. Continuing, our analysis will assess the effectiveness of the exemption under the assumption and evaluate the robustness of our results to it.

## 5 Predictions

We use Machine Learning to predict responses to demanding work-related survey questions that would have been given had the comparison groups been asked them. See section 4 for a list of the questions used. We train and evaluate two models, a random forest and XGBoost, on the respondents group (1), for which the true answer is known. Using the trained models, we make predictions for the individuals in the exempted (2), employed (3), and eligible (4) groups.

In this procedure, it is important to note that sheer prediction performance is not the most important entity. As discussed in Sections 1 and 3, an accurate estimate of the prediction performance, specifically the sensitivity and specificity, is more important, as the proposed methodology can account for misclassification when misclassification rates are known.

### 5.1 Machine Learning models

The machine learning models used are random forest and XGBoost. The main advantage of these models compared to more traditional econometric models is that they are more flexible, as they do not impose a linear relation between the features and outcome variable. We hypothesise that this flexibility leads to a more accurate estimation of  $P(X | Z)$  across regions of  $Z$ , as they are less susceptible to approximation error.

Also, we are only interested in prediction performance, not in the relationship between the features  $Z$  and the outcome variable  $X$ . Therefore, it is not a problem that the ML models are black boxes. Both models are implemented in R.

### 5.2 Features

We construct features from the CBS microdata. As both models are flexible and can generally select important variables themselves, we use many features. In addition to the features already available in the microdata, we construct others ourselves.

For the SPOLIS features, from all monthly entries, only the first month of each year in which the individual appears is selected. To account for differences within each year, new features are

constructed, such as the variation of salary within the year, the variation in hours worked within the year, and the number of times each person appears within the year. Additionally, we use information about the individuals as features, such as gender, highest achieved education level and corresponding education field, and age.

As data from many years is available, both in the microdata and the LISS survey, a target year is chosen for which the outcome is predicted. SPOLIS features are taken at several time windows relative to the target year (e.g., 10 years before, 1 year before, the year of, 1 year after), and age is computed for the target year.

For respondents (1), the target year is the year the survey was completed if the person was working at the time, or the last year the person worked if they were no longer working. For the exempted (2), employed (3), and eligible (4) comparison groups, the target year is set to 2020.

### 5.3 Training and tuning

As the demanding work measures  $X$  are available only for the LISS respondents, the model is trained and evaluated on this sample. To this end, the sample is split into three groups: 52.5% training, 17.5% validation, 30% test. The test set is relatively large, as the emphasis is on obtaining accurate performance estimates. Here, the split is based on individuals rather than responses to ensure no spillover during training/testing. The test set is used only once to avoid overfitting and to produce accurate performance estimates.

Both models produce a score between 0 and 1, corresponding to the estimated likelihood that the response equals 1. As our methodology requires binary estimates, these scores are converted to 0/1 based on a threshold. Increasing this threshold will lead to fewer individuals being classified as 1, thereby increasing specificity and decreasing sensitivity. Reducing the threshold has the opposite effect.

The threshold is chosen that maximises Youden’s  $J$  statistic,

$$J = \text{sensitivity} + \text{specificity} - 1,$$

which is equivalent to maximising the average of the sensitivity and specificity, as discussed in section 3.

Additionally, during training, observations are assigned weights equal to the inverse relative frequency, so that each group ( $Y = 1$  and  $Y = 0$ ) receives the same total weight. Parameters are optimised through grid search based on  $J$ , evaluated on an independent evaluation set. The best model, by  $J$ , per question is chosen to generate the labels for individuals in groups (2) - (4), for whom the true value is unknown.

## 5.4 Estimating the sensitivity and specificity

To estimate misclassification rates, we consider predictions on the test set. Table 3 contains the confusion matrices of the best prediction model for each question. In general, our model seems to have significant explanatory power, as predictions are more often correct than incorrect given the true value. Due to class imbalance, however, oftentimes more than half of the predictions that someone had demanding work are wrong. Response prediction accuracy clearly differs between questions.

| $\hat{X}$      |      |      | $\hat{X}$           |      |      | $\hat{X}$                        |      |      |
|----------------|------|------|---------------------|------|------|----------------------------------|------|------|
| $X$            | 0    | 1    | $X$                 | 0    | 1    | $X$                              | 0    | 1    |
| 0              | 8669 | 3181 | 0                   | 8579 | 3334 | 0                                | 6939 | 3769 |
| 1              | 825  | 1802 | 1                   | 867  | 1697 | 1                                | 1037 | 2732 |
| (a) Profession |      |      | (b) Irregular hours |      |      | (c) Physically demanding (agree) |      |      |

| $\hat{X}$                        |      |      | $\hat{X}$         |      |      | $\hat{X}$         |      |      |
|----------------------------------|------|------|-------------------|------|------|-------------------|------|------|
| $X$                              | 0    | 1    | $X$               | 0    | 1    | $X$               | 0    | 1    |
| 0                                | 8585 | 4243 | 0                 | 9390 | 3884 | 0                 | 8082 | 3945 |
| 1                                | 384  | 1265 | 1                 | 363  | 840  | 1                 | 616  | 1834 |
| (d) Physically demanding (often) |      |      | (e) Heavy lifting |      |      | (f) Kneel / stoop |      |      |

Table 3: Confusion matrices for predictions of questions (a) - (f) on the test set.  $X$  denotes the true value and  $\hat{X}$  denotes predictions.

From the prediction rates, following Eq. 4, the specificity can be estimated by

$$\hat{\theta}_{0|00} = \frac{\ddot{n}_{0,0}}{\ddot{n}_{0,0} + \ddot{n}_{1,0}},$$

and the sensitivity can be estimated by

$$\hat{\theta}_{1|10} = \frac{\ddot{n}_{1,1}}{\ddot{n}_{1,1} + \ddot{n}_{0,1}},$$

where  $\ddot{n}$  is specified as in Table 1. As the sensitivity and specificity differ substantially across different age groups (See Tables A.1 - A.6), we generate two estimates of the sensitivity and specificity, one based on the entire test set and one based on an age-stratified sample of the test set. More specifically, we compute weighted values, where each individual gets a weight equivalent to the fraction of exemption users with the same birth year. The estimated specificities and sensitivities can be found in Table 4, where  $\hat{\theta}_{i|i,all}$  refers to the estimates based on the full test set and  $\hat{\theta}_{i|i,age}$  based on the age-stratified test set.

Table 4 shows that all estimates of the sensitivity and specificity lie between 0.662 and 0.824.

| $Q$ | $\hat{\theta}_{1 1,all}$ | $\hat{\theta}_{0 0,all}$ | $\hat{\theta}_{1 1,age}$ | $\hat{\theta}_{0 0,age}$ |
|-----|--------------------------|--------------------------|--------------------------|--------------------------|
| (a) | 0.686                    | 0.732                    | 0.782                    | 0.685                    |
| (b) | 0.662                    | 0.720                    | 0.670                    | 0.755                    |
| (c) | 0.725                    | 0.648                    | 0.706                    | 0.633                    |
| (d) | 0.767                    | 0.669                    | 0.824                    | 0.710                    |
| (e) | 0.698                    | 0.707                    | 0.712                    | 0.755                    |
| (f) | 0.749                    | 0.672                    | 0.750                    | 0.685                    |

Table 4: Estimates of the sensitivity and specificity, unweighted and age-weighted, for each question (a) - (f).

This suggests that, in line with our observations in Table 3, our model has significant predictive power for both people who had demanding jobs and those who did not. Additionally, it shows that the estimated sensitivities and specificities indeed differ depending on the group on which they are estimated. For most questions (b, d, e, f), both the estimated sensitivity and specificity are higher when evaluated on the age-stratified sample. For question (a) the sensitivity is higher and the specificity is lower, whereas for question (c) they are both lower.

As mentioned in 4, we hypothesise that this age-stratified estimate could be a good approximation for the prediction errors of the eligible comparison group (4) and a better approximation for the exempted group (2), but that they are not correct, as nondifferentiability is not expected to hold.

## 6 Evaluation

To evaluate the RVU levy exemption, we assess what part of exemption users had jobs associated with each aspect of demanding work, and whether they are over- or underrepresented compared to both the employed population and the exemption-eligible population. To this end, we use the predictions and estimated prediction errors of Section 5. This analysis assumes that the nondifferentiability assumption holds. However, as Section 4 suggests it is invalid, we assess the robustness against it.

### 6.1 Does the RVU levy exemption work?

Figure 7 shows the prediction rates for responses to each question associated with demanding work, for each group (1) - (4). We see that for each question, a smaller proportion of each group is assigned demanding work. For questions (b), (c), and (f), the group of exemption users (2) has relatively more predictions for demanding work than the other groups. For (c), the eligible (4) and exempted (2) groups have similar rates, but higher than those of the employed population (3). For (a), the employed group (3) has a significantly higher rate than the other two. However, these figures have not yet corrected for misclassification and correspond to the naive estimate of  $\alpha_y$  defined in Eq. 2.





Figure 7: Prediction rates for each question (a) - (f) for the exempted (2), employed (3), and eligible (4) comparison groups.

To account for misclassification, we compute the corrected estimate  $\hat{\alpha}_y$ , which requires estimates of the sensitivity and specificity. As discussed in Section 5, we have two sets of estimates, one for the overall test set and one for the age-matched test set.

For the sensitivity and specificity estimated on the test set to be exportable to these predictions, we require the nondifferentiability assumption that the misclassification rates are independent of  $Y$ . As discussed previously, this assumption is unlikely to hold for the exempted group (2), but is more likely to hold for the employed and exemption-eligible groups (3) and (4). For these two, we expect  $\theta_{i|i,all}$  and  $\theta_{i|i,age}$ , respectively, could be close to the true values, as both respondents (1), employed (3), and eligible (4) groups are assumed to be representative groups of the same population. Even though the preliminary analysis in Section 4 was not able to confirm this, continuing, we assume these estimates are reliable.

For the exempted group (2), we have little confidence that the sensitivity and specificity are reliable estimates of the true value. Section 6.2 investigates what would have been our conclusion for different estimates of the sensitivity and specificity.

$\alpha$  shows the probability that the response of an exemption user to question  $Q$  would have been 1 or 0. Additionally, we are interested in investigating whether these are over- or underrepresented among exemption users compared to both the full employed population and individuals eligible

to use the exemption. Therefore, we also investigate the odds ratio  $\Psi$ , by estimating

$$\hat{\Psi}_c = \frac{\hat{\alpha}_1(1 - \hat{\alpha}_{0,c})}{\hat{\alpha}_{0,c}(1 - \hat{\alpha}_1)}, \quad (11)$$

where  $c \in \{\text{age}, \text{random}\}$  are the corrected estimates of the employed (3) and eligible (4) groups under the assumption of nondifferentiality with respect to the age-matched test set and full test set, respectively. Estimates of  $\alpha_1$  and  $\Psi$ , under the assumptions above, can be found in Table 5.

| $Q$ | $\hat{\alpha}_{1,\text{all}}$ | $\hat{\alpha}_{1,\text{age}}$ | $\hat{\Psi}_{\text{random}}$ | $\hat{\Psi}_{\text{age}}$ |
|-----|-------------------------------|-------------------------------|------------------------------|---------------------------|
| (a) | 0.23                          | 0.11                          | 0.63                         | 0.23                      |
| (b) | 0.40                          | 0.44                          | 3.32                         | 2.75                      |
| (c) | 0.34                          | 0.34                          | 3.94                         | 4.77                      |
| (d) | 0.25                          | 0.28                          | 0.99                         | 1.24                      |
| (e) | 0.16                          | 0.24                          | 1.65                         | 1.91                      |
| (f) | 0.14                          | 0.17                          | 1.87                         | 3.08                      |

Table 5: Estimates of  $\alpha_1$  and  $\Psi$  for each question (a) - (f).

Table 5 shows our main results.  $\hat{\alpha}_{1,c}$  shows our estimates of  $P(X = 1 \mid Y = 1)$ , the probability that an exemption user would have answered "often" to questions (b), (d), (e), and (f), "agree" or "strongly agree" to question (c), and "supervisory" to question (a), under the assumption of either nondifferentiality of prediction errors between the full test set ( $c = \text{all}$ ) or the age-matched test subset ( $c = \text{age}$ ).

These results indicate that in 2020, of exemption users, around

- (a) 10% - 25% had a supervisory profession,
- (b) 40% - 45% worked irregular hours,
- (c) 30% - 40% had physically demanding jobs,
- (d) 20% - 30% often had physically demanding jobs,
- (e) 15% - 25% needed to lift heavy objects,
- (f) 10% - 20% needed to kneel or stoop.

Additionally, the estimates of the odds ratio suggest that individuals who worked irregular hours (b), had physically demanding jobs (c), needed to lift heavy objects (e) and had to kneel or stoop (f) are overrepresented relative to both the full employed population and the exemption eligible population. Unlike the preliminary analysis suggested, individuals who had a supervisory profession (a) are underrepresented, especially in comparison to their age cohort. Lastly, individuals whose job was often physically demanding (d) appear slightly overrepresented in comparison to their age cohort, but neither under-nor overrepresented compared to the full

unemployed population. This last result is interesting, particularly because of the similarity of questions (c) and (d).

Of course, once again, it should be stressed that these results strongly depend on the assumption of nondifferentiability, of which we argue the validity is unlikely.

## 6.2 Robustness

To investigate the robustness of our results against the nondifferentiability assumption, Figure 8 shows what values the odds ratio would take for different specifications of  $\theta_{0|01}$ , the specificity, and  $\theta_{1|11}$ , the sensitivity.

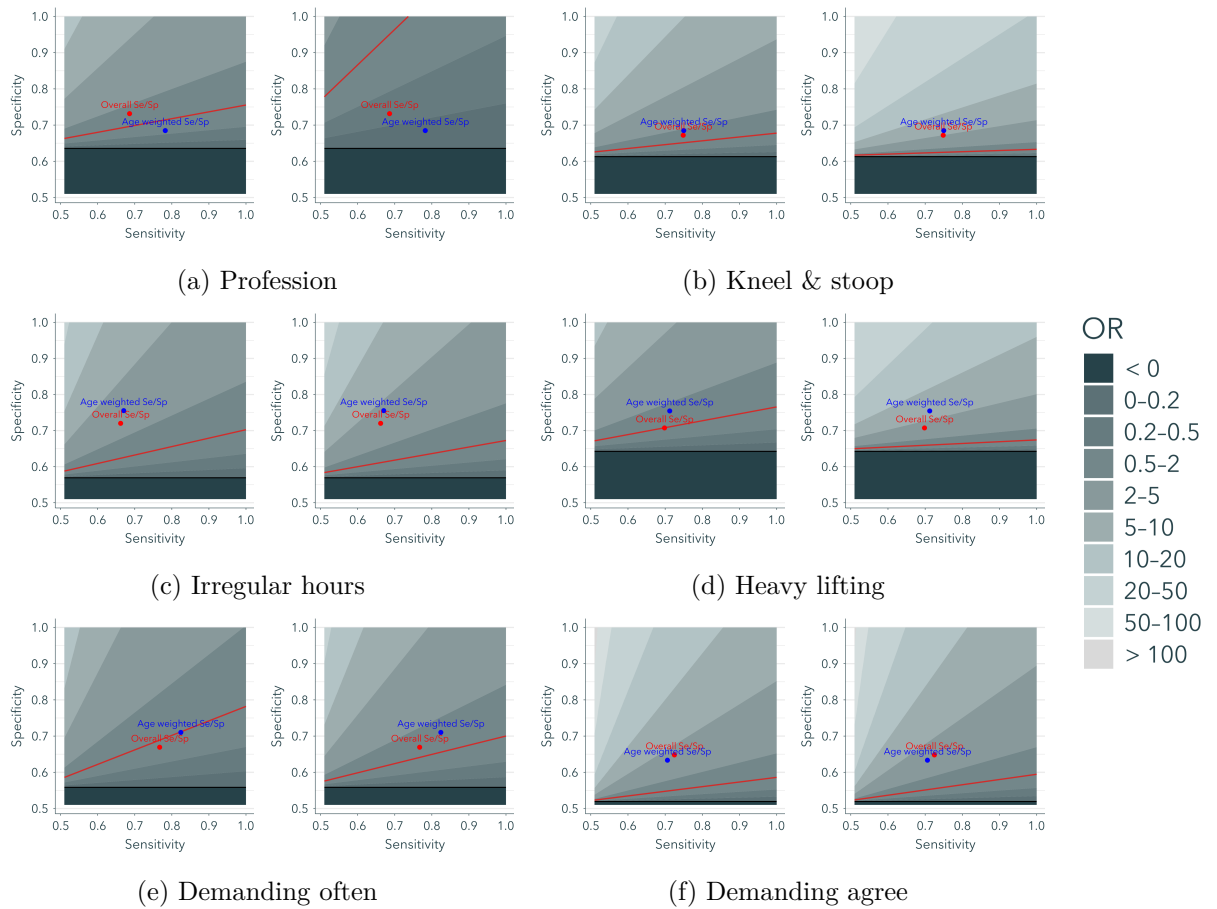


Figure 8: Odds ratios  $\Psi$  for different specifications of the specificity and sensitivity of the misclassified predictions of demanding work. For each pair of figures, the left evaluates the odds ratio against the employed group (3), and the right evaluates the odds ratio against the exemption-eligible group (4). The red and black lines denote the decision boundary of  $\Psi = 1$ , and the unlikely boundary of  $\Psi = 0$ , respectively. The blue and red dots denote the age-weighted and unweighted estimates of  $\theta$ , respectively.

Figure 8 shows the odds ratios for different specifications of  $\theta$ . If the true, unknown, specificity and sensitivity are in the area above the red line, individuals with  $X = 1$  are overrepresented

among exemption users. If the true values are below the red line, they are underrepresented. The combinations for which  $\Psi < 0$  are very unlikely to contain the true values, since these are unlikely to produce the observed prediction rates, as argued in the 70/30 example of Section 3.

The blue and red dots denote the estimated specificity and sensitivity on the age-weighted and unweighted test sets, respectively. These provide some indication of where the true values could lie, but they do not guarantee or provide measurable confidence. However, we hypothesise that the further the estimates are from the decision boundary of  $\Psi = 1$ , the more likely it is that  $X = 1$  is overrepresented among exemption users.

It is apparent that, for some questions, the area associated with either under- and overrepresentation covers close to the full likely set of values. For example, for question (b) to be underrepresented, the true sensitivity and specificity should lie in a small area. Using this argument, we argue that we can be confident in our representation conclusions for questions (c) and (f). For questions (a), (b), and (d), we are confident in our conclusion of representation relative to the exemption-eligible population.

## 7 Conclusion and discussion

This thesis aimed to assess the efficiency of the RVU levy exemption in the period 2021-2025. To this end, we estimated the probability that exemption users had jobs related to several aspects of demanding work. To this end, we predicted for each user what they would have answered to several questions in the LISS Work & Schooling survey, based on features in the CBS microdata. In order to perform inference using these predictions, however, we have to make the unlikely assumption of nondifferentiability of the prediction error with respect to exemption usage.

If this nondifferentiability assumption were valid, a relatively large share of exemption users had jobs associated with demanding work, and individuals who had demanding work are overrepresented among exemption users, both relative to the employed population and the exemption-eligible population. An investigation of the robustness of these conclusions against the nondifferentiability assumption not holding shows that we can be confident in some conclusions, but care is required when making claims.

We conclude that our analysis suggests the RVU levy exemption at least partly reached its intended target, as individuals with demanding work appear to be overrepresented among users. However, for each aspect, a large part of the users did not have jobs associated with it. This conclusion requires careful consideration, as our procedure for predicting survey responses inherently relies on restrictive assumptions unlikely to hold in practice. To make confident claims about the performance of the RVU levy exemption, further analysis and, preferably, more data are required.

Continuing from 2026, the aim is to use RVUs more targeted and controlled. To this end, clearer

guidelines have to be constructed to determine which functions should be eligible for an RVU. As data for RVU use in this period is not yet available, efficiency under the current policy is yet to be evaluated.

## 7.1 Discussion

In addition to the general downsides of using predicted labels discussed throughout this thesis, this study has several severe limitations.

First, the data is not the most reliable. Our measures for demanding work, the LISS survey responses, are self-reported and hence likely to contain mismeasured values. This is not taken into account explicitly in our research. Additionally, the dataset of exemption users appears to contain some noise, as it is likely not a perfect measure of RVU levy exemption usage. This argument is supported by some individuals who took an RVU that did not appear in the employment registration.

Second, next to the widely covered assumption of nondifferentiability of prediction errors between the test set and exemption users, we also assume that misclassification rates are transportable from respondents to the samples employed and exemption-eligible populations, even though a preliminary analysis shows this need not be a valid assumption. This assumption is not explicitly tested and could affect the reliability of our results. Also, we only look at point estimates of the fraction of exemption users with demanding work and do not account for the associated uncertainty.

Third, the discussion of what constitutes demanding work is still ongoing. We proxy for demanding work by using responses to several questions related to different aspects of demanding work, but these are neither perfect nor exhaustive.

Fourth, as the writing procedure began only after access to the CBS microdata was revoked, we encountered several errors during the writing process that could not be recovered. While constructing the comparison samples, I made several mistakes, including stratifying by gender, introducing selection bias by selecting subsamples in the wrong order, and unnecessarily filtering for overlap in the comparison sets. Also, when computing the weighted sensitivity and specificity estimates, the weighting procedure does not work as intended.

## 7.2 Further research

It is evident that there is much room for further research into the RVU levy exemption. This thesis lays a strong foundation in methods for handling missing data and the problems encountered when using predictions, but there are many opportunities for continuation and improvement, starting with evaluating implementation of the RVU from 2026 onward. As this thesis investigates performance of the RVU levy exemption in the period 2021-2025, it does not evaluate the

regulation under the newly proposed changes.

A main problem in this research is the absence of data. Where direct information on whether exemption users had demanding jobs would solve the problem at hand directly, we expect this is unlikely to hold. However, more information from a (small) subsample could already yield further insights, for example, by investigating methods that can work with internal validation samples. In a situation where the data is missing for a random subsample, the assumption of nondifferentiality would be valid and this procedure could be used.

There is much room for improvement in the prediction model. A more thorough preliminary analysis of what features correspond to exemption users could already improve prediction performance. Also, using more informative features could yield improvements. We expect, for example, that health data could significantly reduce misclassification rates. Also, more/other questions regarding demanding work could be used, for example, from the EBB, or by making one indicator for at least one of the questions in this research equal to 1.

In this study, close attention is paid to the fraction of exemption users who had demanding work and to whether they are over- or underrepresented, but the setup of our procedure prevented us from investigating how large a share of the population with demanding jobs used the RVU levy exemption. It could also be of interest to investigate, for example, what fraction of individuals with demanding work used the exemption.

Lastly, the prediction models produced values between 0 and 1, which were converted to binary indicators using a threshold. We hypothesise that in this process, some information was lost, and that using the raw predictions could be a direction for further exploration, perhaps to improve estimates of sensitivity and specificity. Lastly, we hypothesise that, under certain assumptions about prior distributions, the robustness analysis would be an interesting area to investigate from a Bayesian perspective.

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## A Supplementary results

Table A.1: Sensitivity and specificity by birth cohort: `profession`.

| Cohort      | Sensitivity | Specificity |
|-------------|-------------|-------------|
| <1955       | 0.732       | 0.673       |
| 1955        | 0.814       | 0.680       |
| 1960        | 0.785       | 0.732       |
| 1965        | 0.771       | 0.651       |
| 1970        | 0.787       | 0.639       |
| 1975        | 0.665       | 0.678       |
| 1980        | 0.545       | 0.683       |
| 1985        | 0.439       | 0.770       |
| 1990        | 0.565       | 0.870       |
| $\geq 1995$ | 0.165       | 0.936       |

Table A.2: Sensitivity and specificity by birth cohort: `job_extra_hours`.

| Cohort      | Sensitivity | Specificity |
|-------------|-------------|-------------|
| <1955       | 0.616       | 0.637       |
| 1955        | 0.690       | 0.737       |
| 1960        | 0.706       | 0.699       |
| 1965        | 0.778       | 0.675       |
| 1970        | 0.603       | 0.751       |
| 1975        | 0.685       | 0.740       |
| 1980        | 0.597       | 0.664       |
| 1985        | 0.761       | 0.667       |
| 1990        | 0.576       | 0.769       |
| $\geq 1995$ | 0.433       | 0.863       |

Table A.3: Sensitivity and specificity by birth cohort: `job_phys_demand_often`.

| Cohort      | Sensitivity | Specificity |
|-------------|-------------|-------------|
| <1955       | 0.652       | 0.731       |
| 1955        | 0.853       | 0.699       |
| 1960        | 0.897       | 0.631       |
| 1965        | 0.719       | 0.618       |
| 1970        | 0.728       | 0.621       |
| 1975        | 0.766       | 0.666       |
| 1980        | 0.808       | 0.610       |
| 1985        | 0.803       | 0.587       |
| 1990        | 0.750       | 0.714       |
| $\geq 1995$ | 0.479       | 0.821       |

Table A.4: Sensitivity and specificity by birth cohort: `job_phys_demand_agree`.

| Cohort      | Sensitivity | Specificity |
|-------------|-------------|-------------|
| <1955       | 0.645       | 0.626       |
| 1955        | 0.718       | 0.604       |
| 1960        | 0.796       | 0.590       |
| 1965        | 0.719       | 0.567       |
| 1970        | 0.794       | 0.602       |
| 1975        | 0.779       | 0.665       |
| 1980        | 0.740       | 0.610       |
| 1985        | 0.783       | 0.632       |
| 1990        | 0.612       | 0.749       |
| $\geq 1995$ | 0.444       | 0.848       |

Table A.5: Sensitivity and specificity by birth cohort: `job_lift_heavy`.

| Cohort      | Sensitivity | Specificity |
|-------------|-------------|-------------|
| <1955       | 0.500       | 0.810       |
| 1955        | 0.712       | 0.745       |
| 1960        | 0.834       | 0.652       |
| 1965        | 0.713       | 0.642       |
| 1970        | 0.664       | 0.643       |
| 1975        | 0.744       | 0.709       |
| 1980        | 0.716       | 0.700       |
| 1985        | 0.772       | 0.672       |
| 1990        | 0.626       | 0.746       |
| $\geq 1995$ | 0.352       | 0.779       |

Table A.6: Sensitivity and specificity by birth cohort: `job_kneel_stoop`.

| Cohort      | Sensitivity | Specificity |
|-------------|-------------|-------------|
| <1955       | 0.585       | 0.717       |
| 1955        | 0.759       | 0.675       |
| 1960        | 0.865       | 0.638       |
| 1965        | 0.746       | 0.611       |
| 1970        | 0.797       | 0.642       |
| 1975        | 0.828       | 0.678       |
| 1980        | 0.811       | 0.632       |
| 1985        | 0.757       | 0.613       |
| 1990        | 0.628       | 0.736       |
| $\geq 1995$ | 0.443       | 0.784       |